

Demographic Trends in College Graduation Rates

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Project Overview

This project looks at racial gaps in six-year graduation rates across U.S. four-year colleges and universities. The central question guiding this work asks, do certain racial and ethnic groups face larger graduation gaps, and does school selectivity impact those disparities? To answer this question, datasets from the Integrated Postsecondary Education Data System (IPEDS) were used; specifically, institutional characteristics (HD), admissions data (ADM), fall enrollment (EF), and graduation rates (GR). The analysis focuses on three four-year postsecondary institution sectors: public, private nonprofit, and private for-profit, each examining White, Black, Hispanic, and Asian student populations.

Data & Methods

All data were drawn from IPEDS 2024 survey files and merged on the shared UNITID institution identifier. The HD file provided institutional characteristics such as sector, control type, degree level, enrollment size, and locale. The ADM file contributed raw application, admission, and enrollment counts, where acceptance rate and yield rate were computed. The EF file contained fall undergraduate enrollment broken down by race and gender. The GR file provided graduation cohort data of the total cohort denominators and completers within 150% of the normal time as the numerator.

Additionally, flag columns prefixed with X were dropped from all files before merging. Graduation rates were computed independently for each racial group by dividing that group's completers by their own cohort size. It was not computed by the total institutional cohort in order for the group-level rates to reflect actual within-group completion. All rates were scaled to [0, 1], and infinite values were replaced with NaN. A gender gap variable was also calculated as the difference between female and male graduation rates, where a positive value indicates women graduate at a higher rate.

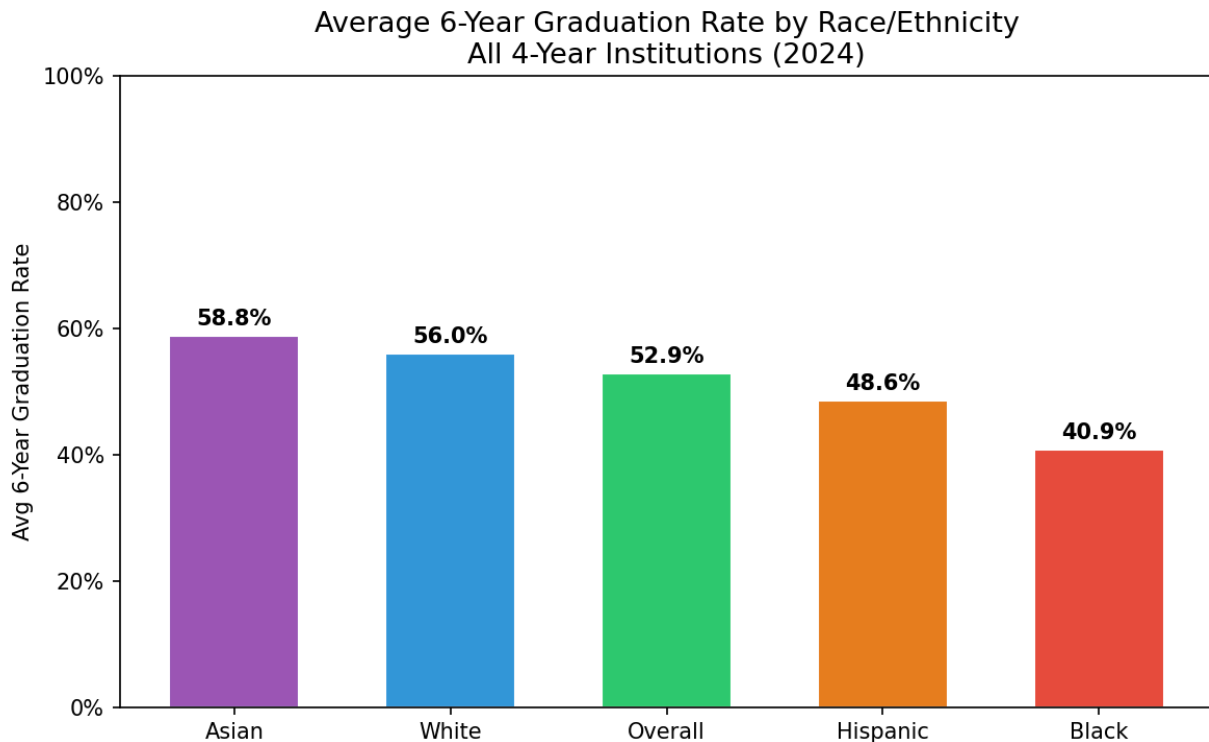
Finally, the final merged dataset included all four-year institutions across the three primary sectors. Schools were further classified by selectivity using acceptance rate, separated into Highly Selective (under 30%), Selective (30–60%), and Open/Less Selective (above 60%). Selectivity-based comparisons were only for institutions reporting both admissions and graduation data. Race/ethnicity enrollment proportions were calculated by each group's count divided by total undergraduate enrollment.

Findings

Overall, we found that, averaged across all four-year institutions, Asian students graduate at the highest rate (58.8%), followed by White students (56.0%), Hispanic students (48.6%), Black students (40.9%), and the overall institutional average (52.9%). The Black and White gap

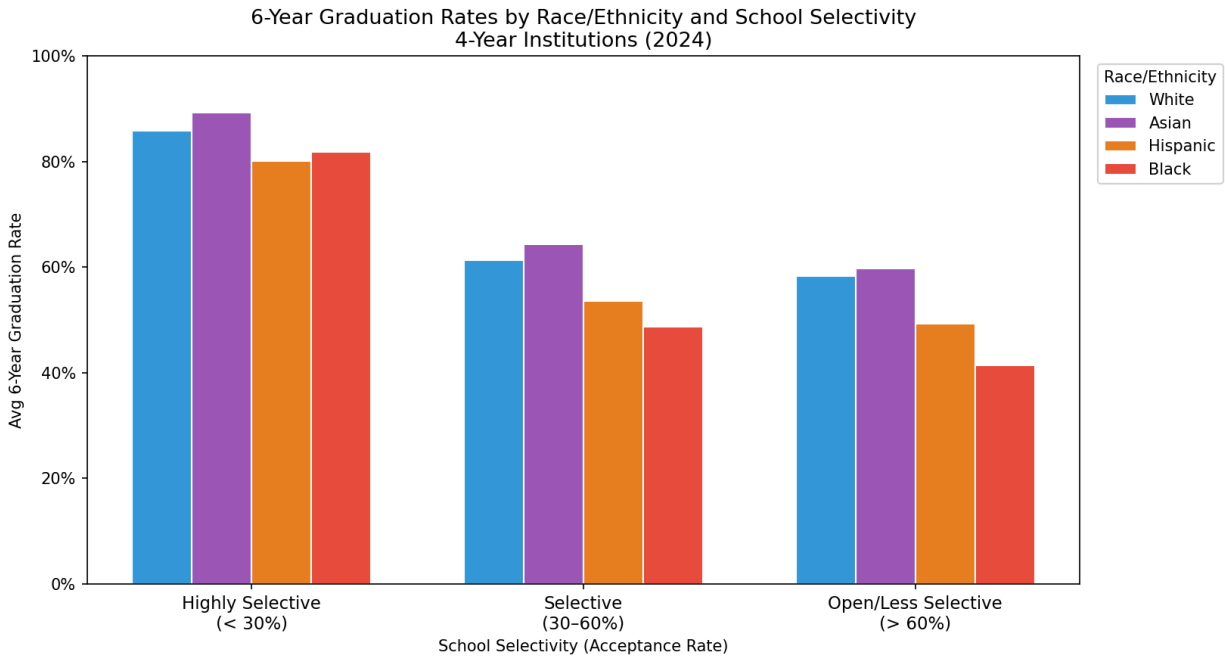
of approximately 15 percentage points is the largest disparity in the data, and both Black and Hispanic students fall meaningfully below the overall national average.

Figure I.



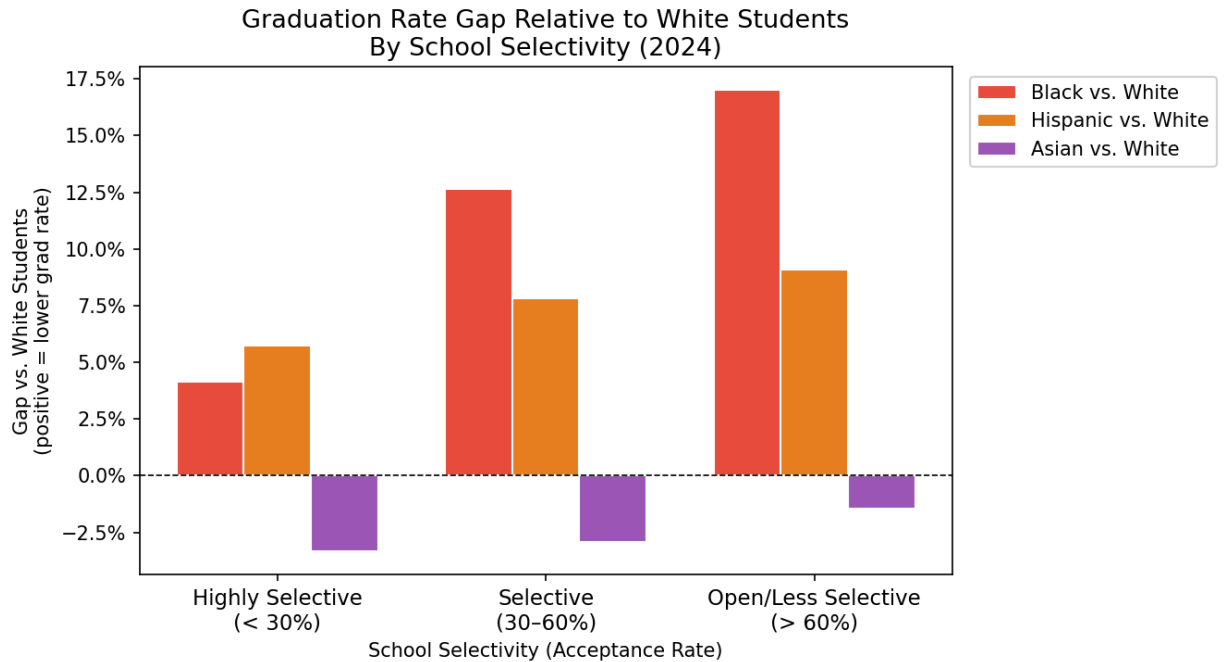
This bar chart illustrates how breaking down graduation rates by selectivity tier shows that outcomes improve at more selective schools across all groups. At highly selective institutions, rates are toward the mid-to-high 80s; Asian students at roughly 89%, White at 86%, Black at 82%, and Hispanic at 80%. The gaps between groups are narrower at this tier than anywhere else in the data.

Figure II.



This bar chart shows that selective schools' rates drop to the low/mid 60s for White and Asian students and into the high 40s and low 50s for Black and Hispanic students. At open or less-selective schools, White and Asian students remain near 58–60%, while Hispanic students fall to around 49% and Black students to roughly 41%. This is almost identical to their national averages, which makes sense given that open-access schools make up the bulk of the dataset.

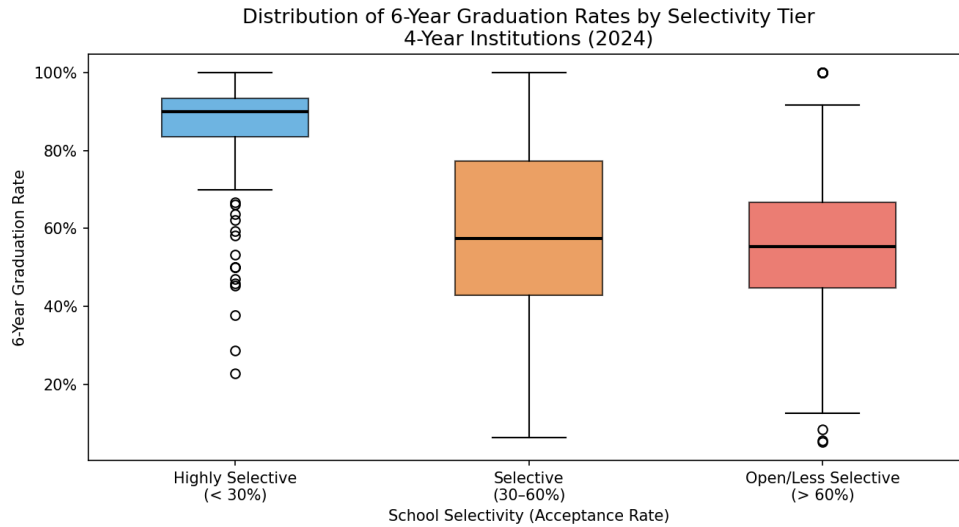
Figure III.



The bar chart illustrates each group's gap relative to White students within the same selectivity tier. The Black and White gap grows from roughly 4 percentage points at highly selective schools to 12.5 points at selective schools and nearly 17 points at open institutions. The Hispanic and White gap follows a similar pattern, from about 6 points to 9 points across the same tiers. Asian students outperform White students at every tier, shown by the consistently negative bars, meaning the gap for Asian students is in the opposite direction.

This suggests that the racial graduation gap depends on the type of institution a student attends. At highly selective schools, the combination of institutional resources, peer environments, financial aid capacity, and stronger student preparation for school all play a role. At less selective schools, these factors weaken, and inequities in academic preparation, financial stability, and access to support services go directly into differential completion rates. Black students at highly selective schools graduate at 82%, only 4 points behind White students, while that gap widens to 17 points at open-access schools. This suggests that the gap is less about racial group differences and more about the institutional conditions students are placed in.

Figure IV.



As illustrated by the box plot data, highly selective schools are clustered tightly near the top. Not only do they produce high average graduation rates, but there is very little variation among them. This means that almost all students at these institutions complete their degrees regardless of which specific school they attend. However, at open-access schools, these outcomes are spread across the full range, all the way from nearly zero to near-perfect graduation rates. This spread means that attending an open-access school is a less predictable experience, and the students most at risk are in the tier with the least consistency.

Figure V.



The scatter plot confirms this relationship at the institution level. There is a clear negative correlation between acceptance rate and overall graduation rate. The schools that admit a smaller number of applicants consistently produce higher graduation rates across all three sectors. Private nonprofit institutions cluster in the upper-left with low acceptance rates and high graduation rates. Public four-year schools are more distributed in the middle range of both axes. Private for-profit schools appear throughout, but have lower graduation rates regardless of selectivity. Among open-access schools, outcomes vary as some achieve strong graduation rates despite near-universal admission, while others fall below 20%, demonstrating wide variation in institutional quality and support.

Limitations

One limitation is that the graduation rates are averaged at the institution level without accounting for enrollment size; a school with 500 students has the same influence on the average as one with 40,000. Enrollment-weighted averages would more accurately reflect the national student experience. Furthermore, selectivity is measured using the acceptance rate alone. However, incorporating SAT/ACT score percentiles would allow for a more accurate selectivity

index. The dataset also includes locale (urban/rural), state, and institution size variables that could support geographic analysis that is not present here.

The observed gaps are descriptive and not causal. Graduation rate disparities at less selective institutions may reflect a mix of financial pressures, such as the fact that students need to work while enrolled, first-generation college status, and differences in pre-college academic preparation. All of this is correlated with both race and institution type, but not directly controlled in this analysis. The enrollment composition variables from the EF file could also be used in future regression synthesis to account for these effects.

UNCF Research Support

Additionally, research from UNCF supports the graduation rate disparities we found in our analysis. Only 45.9% of Black students enrolled in four-year public institutions complete their degrees within six years, which is the lowest rate of any racial or ethnic group, and black men are even lower at 40%. This aligns with our findings from the IPEDS GR2024 data, which showed consistent graduation rate gaps across demographic groups at the national level.

Conclusion

Overall, data trends suggest that Black and Hispanic students graduate from four-year institutions at substantially lower rates than White and Asian students, with the Black and White gap sitting at about 15 percentage points nationally. That gap narrows around 4 points at highly selective schools and widens to 17 points at open-access schools. This demonstrates that the disparity isn't fixed and that it shifts depending on where students end up. This raises the question of who actually gets access to the kinds of schools where these gaps are smaller. Highly selective institutions have the resources, peer environments, and financial aid capacity that seem to help level the playing field, but most students aren't attending those schools. For students at open-access institutions, the experience is less predictable due to a lack of resources, as outcomes range from near-zero to full graduation rates depending on the school. It is important to consider that open-access schools do close the gap. That means selectivity alone isn't the deciding factor. Factors such as institutional support, financial resources, and how well a school serves students who face more barriers all matter too. Future work should look into what those schools are doing differently, because if some less selective institutions can achieve more equal outcomes, there's no reason others can't either.